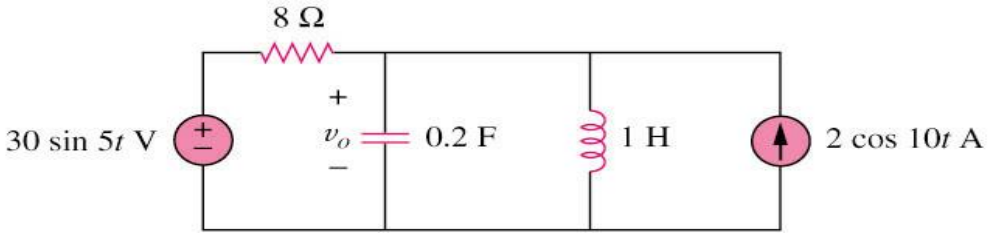
ELEC 226 HOMEWORK-1

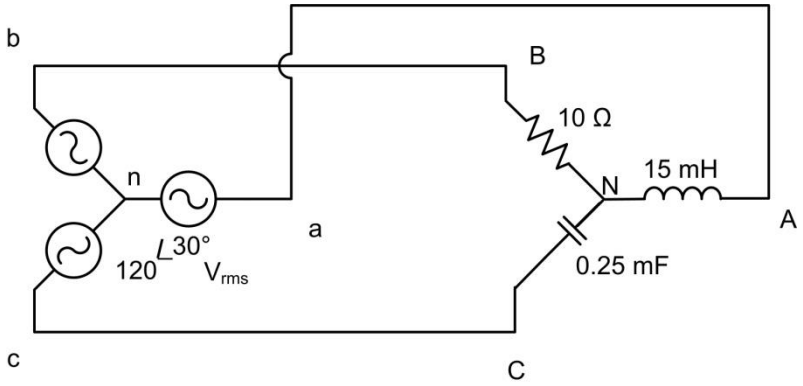
Q-1) Calculate v_0 voltage in the circuit by using Superposition Theorem. (Use the given notation for v_0 voltage).



Q-2) For the 3-ph. system given below, assume a balanced sourcewith a positive phase sequence.If the operating frequency is 50 Hz

a) Find the magnitude of V_{AN}, V_{BN}, V_{CN} voltages

b) Find the total complex power suppiled by the sources.



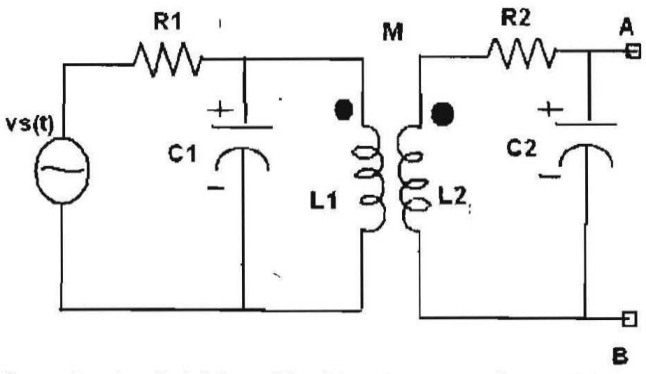
Q-3)

For the circuit, unknown values are: V_s(t)=100 cos(5t) ,$R_1 = 12\ \Omega$, $R_2 = 5\ \Omega$, $L_1 = 0.3\ H$,$L_2 = 1.2\ H$,$C_1 = C_2 = \frac{1}{100}\ F$ and $M = 0.6\ H$.

a)Calculate the Thévenin Equivalent Circuit between A-B port.

b)What is the impedance value which satisfy the max. power between A-B.

c)For the impedance value found in b, calculate the apparent power.



Q-4)

The source $V(t) = 230\angle 0°$ feeds 3 parallel Loads. For the loads below

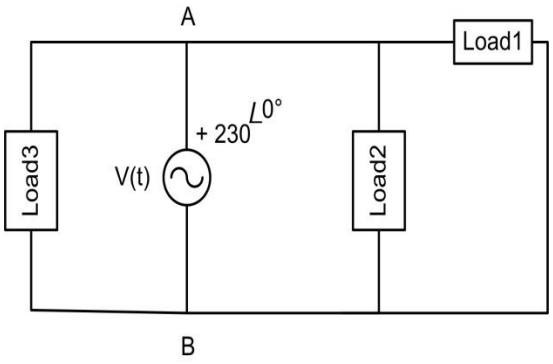
Load 1 : S=25 kVA, PF=0.6 inductive,

Load 2 :P=5 kW, PF=0.8 capacitive

Load 3 :P=10 kW, PF=1

a) Calculate the condansator value to set the power factor at 0.95. (Freq.=50 Hz)

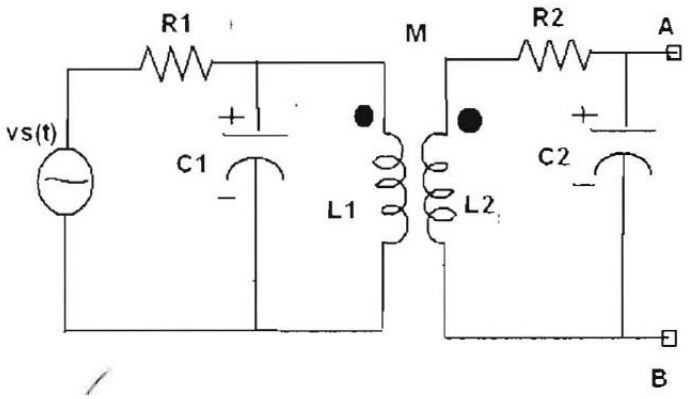
b) Find the current value before and after the compansation. Explain the difference, if it exist.



Q-5) Voltage source Vs(t)=100.Cos(5t) and R1=12 ohm, R2=5 ohm, L1=0.3h, L2=1.2H, C1=C2=1/100 F and mutual inductance (M) is 0.6 H.

A) Please find thevenin equivalent between A and B.

B) To draw maximum power from voltage source, please find the impedance that must be connected between A and B.



Q-6) The source whose voltage is $230\ V$ is providing power to 3 loads which are Yük-1, Yük-2 and Yük-3:

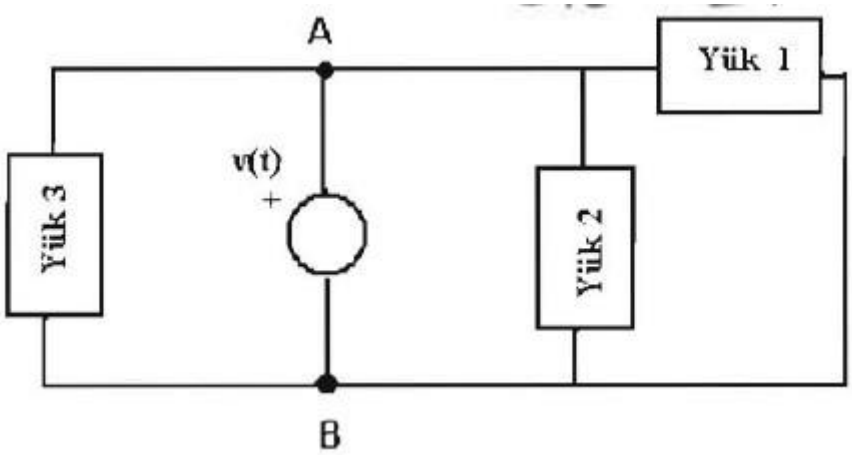
Yük 1: S= 20 kVA, pf=0.6 (inductive)

Yük 2: P= 10 kW, pf=0.8 (capacitive)

Yük 3: P= 10 kW, pf=0.6 (inductive)

A) Please find the capacity value of the capacitor to make power factor 0.95 (f=50hz). The location of the capacity must be in between A and B points.

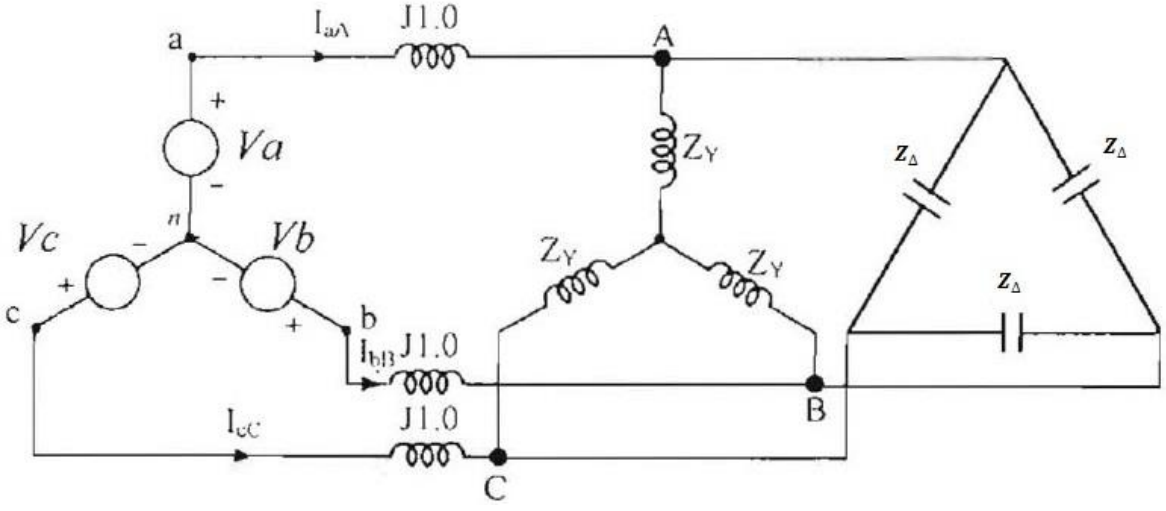
B) Please find the current given by source before and after compensation.



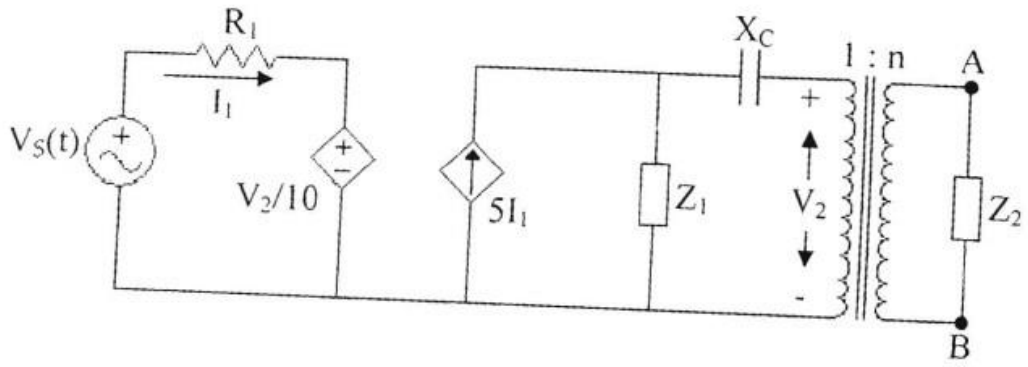
Q-7) Phase voltages are $V_a = 220\angle 30^\circ$, $V_b = 220\angle 150^\circ$ and $V_a = 220\angle -90^\circ$. The loads information: $Z_\Delta = -j100ohm$ and $Z_\gamma = j100ohm$.

A) Please find IaA, IbB and IcC.

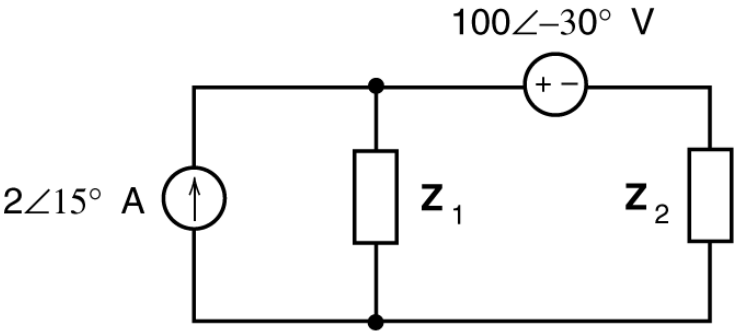
B) Please find total active, reactive and apperant power consumed by loads.



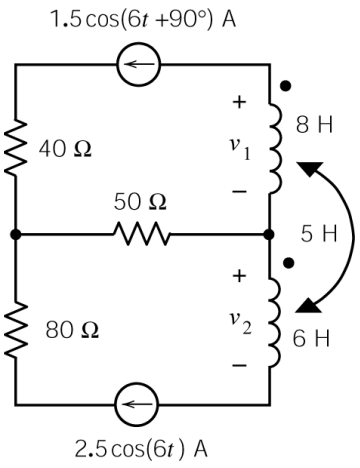
Q-8) To draw maximum power from the circuit, please find Z_2 value. $V_s(t) = 20\cos 100t$, $R_l = 20ohm, Z_l = 10 + 10j, X_c = -j$, n=5.



Q-9) The current source the circuit shown in Figure below supplies 131.16 – *j*36.048 VA and the voltage source supplies 64.2275 – 87.8481VA. Determine the values of the impedances Z1 and Z2.



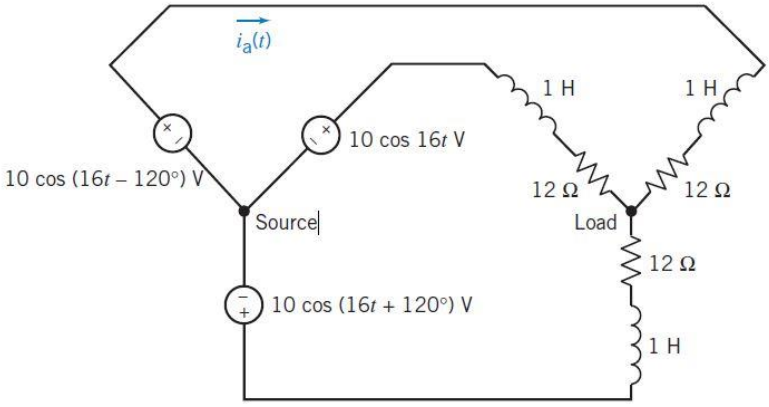
Q-10) The circuit shown in Figure below is represented in the time domain. Determine coil voltages *v*1 and *v*2.



Q-11) Wye-connected source and load are shown in Figure below.

(a) Determine the rms value of the current *i*A (*t*).

(b) Determine the average power delivered to the load.



Q-12) The circuit in Figure below consists of a source connected to a load.

(a) Suppose *R* = 9 Ω and *L* = 5 H. Determine the average, complex, and reactive powers delivered by the source to the load.

(b) Suppose *R* = 15 Ω and *L* = 3 H. Determine the average, complex, and reactive powers delivered by the source to the load.

(c) Suppose the source delivers 8.47 + *j*14.12 VA to the load. Determine the values of the resistance, *R*, and the inductance, *L*.

(d) Suppose the source delivers 14.12 + *j*8.47 VA to the load. Determine the values of the resistance, *R*, and the inductance, *L*.

